

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

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ABSTRACT

A frozen CLIP region can pass locality and mean-foil screens while responding more strongly to a non-target foil. We audit the maximum-target-drop single region from CCI candidates using matched prompt-mean and aggregate worst-foil margins. On COCO/OpenAI B/16, 41.18% of eligible images pass both screens yet have negative worst-foil margin; 64.38% [63.67,65.07]% of screen-passers fail. Selection-stage target-tolerant reranking at $\epsilon = .02$ switches 2.0–2.6% of images and improves normalized aggregate margin by .0150/.0108/.0118/.0087 across COCO/VOC and B/16/B/32. All specificity-gain 10,000-resample paired confidence intervals exceed zero; held-out target-response intervals include zero. Gains persist with category-disjoint and annotation-absent foils. Margin improvement differs from sign repair: COCO B/16 negative-margin prevalence falls only from 74.04% to 73.89%, while switched-image bbox precision falls by 4.95 percentage points on average. Reranking demonstrates sparse local margin improvement within frozen candidates, not broad failure elimination or loss-free explanations.

Index Terms— CLIP, explainability, interventions, model auditing, worst-foil specificity

1 INTRODUCTION

Frozen vision-language models such as CLIP [1] are often interpreted by ranking image regions according to the change in image–text similarity after an intervention. CCI [2], for example, clusters CLIP patch tokens into regions and combines cluster importance into a heatmap. We audit only its candidate region with maximum target-score drop (CCI-top1), not the complete heatmap. A large drop says that a region matters for the target; it does not establish that the effect is *specific* to the target. Removing generic object evidence can reduce a competing semantic foil even more, while bounding-box overlap remains high. Likewise, averaging over many easy foils can hide one high-impact confounder. On COCO B/16, 41.18% of eligible images have bbox precision $\geq .5$ and positive aggregate mean-class margin, yet negative worst-foil

margin. Among the 17,725 screen-passers, 11,411 (64.38%) fail the worst-foil audit.

We audit the declared foil-response vector of the *selected* frozen region and report target response, worst-foil specificity, and spatial locality separately. The contribution is the post-selection audit and controlled evidence of local margin improvement, not the max/CVaR operator itself [3]. COCOA [4] defines a corpus-versus-foil attribution objective; CASE [5] diagnoses class-insensitive saliency and builds contrastive explanations. Here the object is a fixed selected region’s intervention-response vector: we test what locality and mean-class screens miss, then probe improvement with selection separated from held-out audit. Related CLIP attribution [6, 7, 8], class/concept contrasts [9, 10, 11], and caption foils [12] motivate this distinction. Image Description Enhanced CLIP-Adapter (IDEA) [13] combines visual features with image descriptions for few-shot classification. Our audit instead examines target specificity of selected regions with the model held fixed.

Reranking is a controlled probe of local improvement, not the principal algorithmic claim. Negative relative specificity does not imply that the region provides no faithful evidence for the target: evidence may support several classes.

2 WORST-FOIL AUDIT AND TARGET-TOLERANT RERANKING

For checkpoint m , $s_m(x, c) = \tau_m \langle z_m(x), e_m(c) \rangle$ uses unit-normalized embeddings and the checkpoint’s learned $\tau_m = \exp(\text{logit_scale}_m)$, before softmax. For candidate region R and concept prompt c , define

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c), \quad (1)$$

where $\mathcal{I}$ is a frozen internal intervention. Our single-region CCI baseline selects

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t), \quad (2)$$

using only the selection prompt. CCI-top1 denotes this baseline; the CCI subscripts and figure labels retain this shorthand.

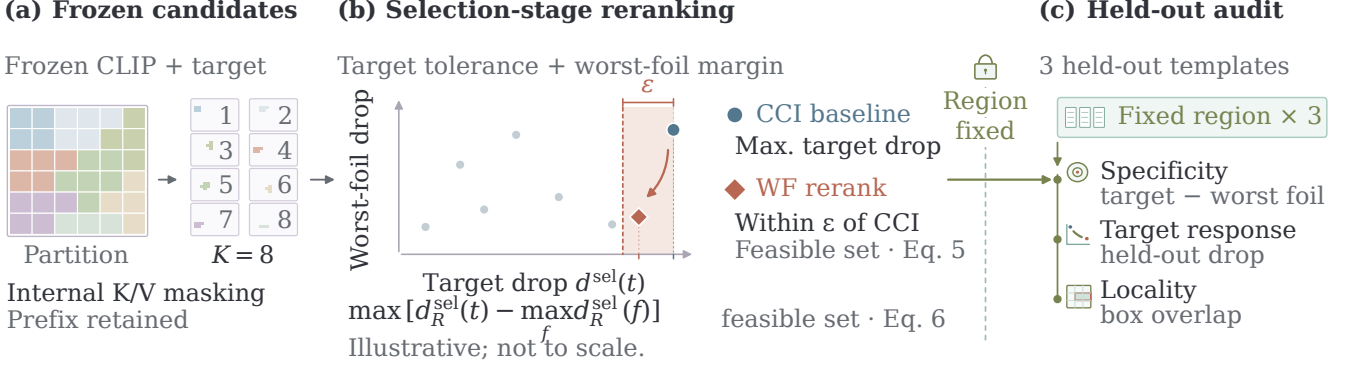


Figure 1: Selection and held-out audit. Frozen CCI candidates are selected or reranked only from selection-stage tensors. The region is fixed before held-out specificity, target-response, and locality evaluation. The candidate-response diagram is illustrative and not to scale.

For non-target foils $\mathcal{F}_t$, a worst-foil margin compares the target response with the strongest foil response.

Matched held-out estimands. Selection uses “a photo of a {category}”. Held-out evaluation uses “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. Let $e_h(c)$ denote the stored unit-norm text embedding for held-out prompt h , and $n_c = \|H^{-1} \sum_h e_h(c)\|_2$. We apply the same class-specific normalization rule, $\tilde{d}_h(c) = d_h(c)/n_c$ and define

$$P_{\max}^{\text{pm}}(R, t) = H^{-1} \sum_h [\tilde{d}_h(t) - \max_f \tilde{d}_h(f)],$$

$$P_{\max}^{\text{agg}}(R, t) = H^{-1} \sum_h \tilde{d}_h(t) - \max_f H^{-1} \sum_h \tilde{d}_h(f). \quad (3)$$

Thus $P_{\max}^{\text{pm}} \leq P_{\max}^{\text{agg}}$: the former permits a different hardest foil per template. Both are ontology-relative specificity measures. For the mean-class screen, write $\bar{d}_c = H^{-1} \sum_h \tilde{d}_h(c)$. The archived screen uses $P_{\text{mean,all}} = \bar{d}_t - |\mathcal{C}|^{-1} \sum_{c \in \mathcal{C}} \bar{d}_c$ over all dataset classes, equally weighted, on these same held-out responses. Excluding the target gives $P_{\text{mean,foil}}$, with $P_{\text{mean,all}} = (|\mathcal{C}| - 1)P_{\text{mean,foil}}/|\mathcal{C}|$; their positive screens coincide. This class average differs from prompt-mean.

Selection-stage target tolerance. For each candidate define

$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_t} d_R^{\text{sel}}(f). \quad (4)$$

Given tolerance ϵ , form

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

and rerank only this feasible set:

$$R_{\text{WF}}(\epsilon) = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

Because $R_{\text{CCI}} \in \mathcal{R}_\epsilon$, the feasible set is nonempty; any switch loses at most ϵ of the *selection-stage* target drop. This is not a held-out target guarantee. Equation 6 reads only selection-stage tensors, and held-out prompts are used only after the

region is fixed. We use $\epsilon = .02$ as a fixed operating point, not an optimized value. Its cosine-scale tolerance is $.02/\tau_m$, not necessarily equal across checkpoints. Figure 1 summarizes the separation.

3 EXPERIMENTAL PROTOCOL

Models, intervention, and endpoints. We use $K = 8$ frozen CCI regions on COCO val2014 [14] (27,708 eligible images) and VOC2007 test [15] (1,943), with OpenAI CLIP B/16 and B/32. Reranking changes only which frozen candidate is selected. Each image has one target: the largest-area annotated instance’s category (COCO annotation area; VOC box area). Eligibility requires another annotated category and target-instance area between 2% and 80% of the image; COCO crowd annotations are excluded. Full foils comprise every other dataset category (79 for COCO, 19 for VOC). The intervention blocks selected patch key/value columns in every visual-transformer block and head by adding $-\infty$ before attention softmax, while retaining the prefix token. Cached patch features are clustered by K-means ($n_{\text{init}} = 3$, 50 iterations, seed $1701 + i$ for cache index i). BBox precision is the fraction of selected patch centers in the union of transformed boxes of the target category, not only its largest instance, and is reported separately because perturbation metrics depend on the intervention operator [16]. Ties use the first exact NumPy argmax.

Held-out controls and uncertainty. Selection uses one template and evaluation uses the three held-out templates in Eq. 3. Cross-foil controls use ten fixed seeds 1701, 2701, $\dots$, 10701 to permute categories into equal alternating folds. Each direction selects using one fold and audits on the other, excluding the target from both. The 20 seed–direction deltas are averaged within each image before paired unique-image resampling; annotation-absent evaluation removes categories annotated as present without claiming semantic absence. Fixed 5/19-foil subsets probe cardinality and composition jointly. Intervals use 10,000 paired unique-image bootstrap resamples; the archive fixes the base seed at 1701 and records deterministic

Table 1: CCI-top1 vs. WF at $\epsilon = .02$. $P = P_{\max}^{\text{agg}}$; brackets are paired 95% CIs; “Neg.” is baseline→WF negative-margin rate.

Setting	P_{CCI}	ΔP	Δd_t	Δbbox	Sw.	Neg.
COCO B/16	-.342	+0.0150 [+.0131,+0.0169]	+0.00005 [-.00034,+0.00044]	-.00118 [-.00199,-.00036]	2.39%	74.04→73.89%
COCO B/32	-.304	+0.0108 [+.00945,+0.0122]	-.00010 [-.00046,+0.00027]	-.00080 [-.00162,+0.00002]	2.21%	73.91→73.75%
VOC B/16	+.168	+0.0118 [+.00677,+0.0183]	-.00085 [-.00227,+0.00059]	-.00340 [-.00652,-.00034]	2.57%	58.16→57.59%
VOC B/32	+.186	+0.00867 [+.00441,+0.0143]	-.00069 [-.00184,+0.00040]	+0.00023 [-.00288,+0.00341]	1.96%	55.89→55.58%

setting/metric-specific seeds. Only four provenance-valid OpenAI settings enter the direct reranking analysis; two historical LAION settings are excluded because their text-normalization provenance is unresolved.

4 RESULTS

Table 2: COCO B/16 screen audit. A : $\text{bbox} \geq .5$ and mean-class margin > 0 ; F : worst-foil margin < 0 . Image-bootstrap 95% CIs.

Event	Count	Rate [95% CI], %
A	17,725/27,708	63.97 [63.41,64.53]
F	20,516/27,708	74.04 [73.53,74.56]
$A \cap F$	11,411/27,708	41.18 [40.62,41.76]
$F A$	11,411/17,725	64.38 [63.67,65.07]

Worst-foil debt. On COCO/OpenAI B/16, CCI-top1 has normalized $P_{\max}^{\text{agg}} = -.342$ [-.357,-.327], and 74.04% [73.56,74.61]% of selected regions have negative aggregate margin. Its held-out raw target response is .939 [.924,.953] and bbox precision .679 [.674,.683]. These are full-sample means, not estimates within the negative-margin subset. Table 2 decomposes the screen failure. Within the negative-margin subset, mean raw target drop is .481 and 90.40% have positive target drop.

Sparse margin improvement. Table 1 gives the direct baseline comparison. Normalized aggregate and prompt-mean deltas are positive with CIs above zero in all four settings; the corresponding prompt-mean deltas are +.0150, +.0108, +.0119, and +.00867. Held-out target-response intervals include zero in all four; this does not establish equivalence or a no-loss guarantee. Locality is not uniformly preserved: B/16 on COCO and VOC shows small bbox decreases, while B/32 intervals include zero.

The gain is not merely selector–endpoint reuse. Figure 2 shows that the gain persists under category-disjoint cross-foil and annotation-absent evaluation. Pooled normalized aggregate gains are +.01052 [+.00901,+0.01198], +.00728 [+.00629,+0.00846], +.00705 [+.00296,+0.01140], and +.00438 [+.00131,+0.00784] for COCO B/16, COCO B/32, VOC B/16, and VOC B/32. Restricting evaluation further to annotation-absent foils remains positive on the full eligible image population: +.00939, +.00697, +.00809, and +.00443, with all paired

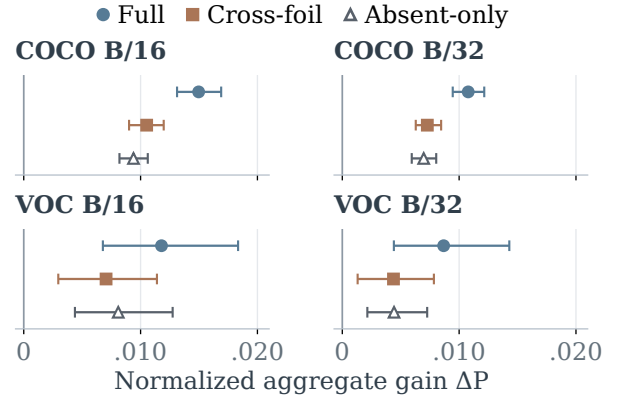


Figure 2: Held-out normalized aggregate gains at $\epsilon = .02$ for full, category-disjoint cross-foil, and annotation-absent evaluation; bars are paired 95% CIs.

CIs above zero. These controls support gain persistence, not baseline failure-rate persistence under the same restrictions.

Descriptive same-feasible-set comparisons. In Table 3, WF has larger point estimates than uniform, target runner-up, and mean-foil selection; Max-.1 nearly matches WF. These are gains relative to CCI-top1, not paired WF-minus-alternative CIs, so they do not establish statistically significant superiority.

Why switching is sparse. At $\epsilon = .02$, COCO B/16 has 26,405 singleton feasible sets; 1,215/80/8 images have 2/3/4 candidates. Only 1,303/27,708 (4.70%) permit an alternative, and 663/1,303 (50.88%) switch.

The effect also reduces the breadth of foil violations. As a secondary held-out diagnostic, let $\bar{d}(c) = H^{-1} \sum_h \bar{d}_h(c)$ for the already fixed region. We report V_{rate} , the fraction of foils with $\bar{d}(f) > \bar{d}(t)$, and $V_{\text{mass}} = |\mathcal{F}_t|^{-1} \sum_{f \in \mathcal{F}_t} [\bar{d}(f) - \bar{d}(t)]_+$, averaging over all foils, including zeros. On COCO/OpenAI B/16, reranking at $\epsilon = .02$ changes V_{rate} from .18973 to .18751 and V_{mass} from .05878 to .05708. Violation breadth therefore also decreases descriptively.

Foil-cardinality sensitivity. Adding foils can increase the maximum response, but the fixed subsets in Table 4 vary cardinality and composition jointly. The reranking gain stays positive at both matched counts, so the effect is not created solely by evaluating against a larger foil set.

Table 3: Same-feasible-set baselines at $\epsilon = .02$; normalized aggregate $\Delta P_{\max}$. Max-.1: $d_t - .1 \max_f d_f$.

Setting	WF	Max-.1	Mean	Unif.	2nd d_t
COCO B/16	.0150	.0147	.0025	.0006	.0002
COCO B/32	.0108	.0106	.0015	-.0025	-.0023
VOC B/16	.0118	.0113	.0073	.0010	.0026
VOC B/32	.0087	.0083	.0055	-.0044	-.0046

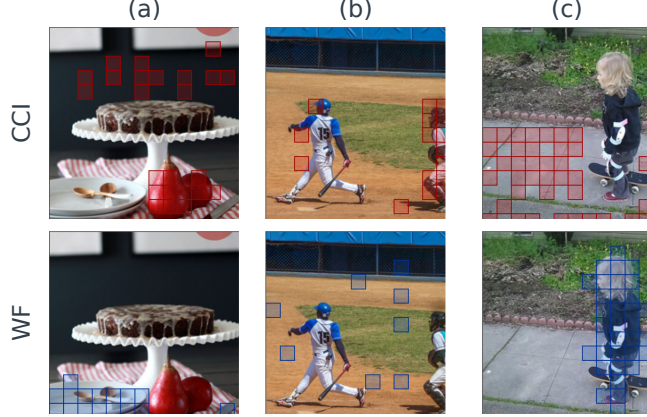


Figure 3: Archived rule-selected COCO/OpenAI B/16 cases at $\epsilon = .02$ (top/bottom: CCI/WF); ΔP is normalized aggregate gain: (a) locality gain, dining table, $\Delta P = +4.58$; (b) specificity–locality trade-off, person, $\Delta P = +0.78$; (c) counterexample, person, $\Delta P = -0.98$. Bbox precision: $.46 \rightarrow 1.00$, $.81 \rightarrow .11$, and $.02 \rightarrow .88$, respectively.

Table 4: COCO foil-count sensitivity. “Neg.” is baseline negative-margin rate; ΔP is normalized aggregate gain.

Setting	Neg.@5	Neg.@19	Neg.@all	ΔP_5	ΔP_{19}
COCO B/16	45.8%	63.5%	74.0%	+.00872	+.01074
COCO B/32	45.0%	62.8%	73.9%	+.00841	+.00947

Cross-template consistency. Across the three held-out phrases, mean pairwise hardest-foil identity agreement is 59.80%. All three agree on the exact category in 46.36% of images and on its stored semantic family in 53.69%; the latter two use the same all-three criterion, not the pairwise criterion. Families follow the frozen metadata mapping, e.g., vehicle and animal.

Tolerance sensitivity. Table 5 uses the same frozen candidates and selection tensors; $\epsilon = .02$ was fixed rather than selected from it. As tolerance grows, switches increase while $\Delta P \mid \text{sw}$ stays stable. Full-sample target/bbox deltas are $(+.00005, -.00118)$ at $.02$, $(-.00049, -.00223)$ at $.05$, and $(-.01396, -.01627)$ at $.20$, exposing the target–specificity trade-off.

Switch-conditioned outcomes. Table 6 uses $E[\Delta q \mid \text{sw}] = E[\Delta q] / \Pr(\text{sw})$ on unrounded archived statistics. VOC B/16 loses 13.20 bbox percentage points among switches. For COCO B/16, switched-image bootstrap gives mean $\Delta \text{bbox} -0.0495$ $[-.0842, -.0157]$ and raw $\Delta d_t = +.00218$ $[-.01364, +.01783]$. Median bbox change is 0 (quartiles

Table 5: COCO/B16 tolerance sensitivity; ΔP is full-sample gain, while $\Delta P \mid \text{sw}$ and $\Pr(+)$ condition on switches.

ϵ	Sw.	ΔP	$\Delta P \mid \text{sw}$	$\Pr(+)$
.01	1.23%	+.00791	+.642	90.9%
.02	2.39%	+.01497	+.626	92.2%
.05	6.24%	+.03906	+.626	91.6%
.10	11.68%	+.07643	+.654	92.0%
.20	21.87%	+.14479	+.662	92.9%

$-.3242, +.1137$); 284/663 switches improve margin but lose locality. Only 42/663 repair the sign.

Table 6: Switch-conditioned outcomes at $\epsilon = .02$. Means use unrounded archive ratios; Gain/Repair are margin-improvement/sign-repair percentages. Conditional CIs for other settings are unavailable.

Setting	n_{sw}	Δd_t	Δbbox	Gain/Repair
COCO B/16	663	+.00218	-.04950	92.2/6.33
COCO B/32	613	-.00457	-.03619	86.6/7.83
VOC B/16	50	-.03228	-.13199	92.0/22.0
VOC B/32	38	-.03486	+.01163	92.1/15.79

Cases and uncertainty. Figure 3 selects an annotation-absent-foil gain, a case near the switched-gain 75th percentile, and the worst switched loss; it does not estimate frequencies. Added screen intervals use 10,000 image-bootstrap draws (seed 1701); conditional mean intervals resample the 663 switched images (seed 1702).

Interpretation and scope. Inference concerns eligible multi-category images under fixed templates, foils, candidates and intervention; bootstrap does not establish robustness to new configurations. Missing annotations need not imply semantic absence. The screen decomposition is verified for COCO B/16; other settings and restricted-foil baseline conditional rates remain unreported.

Conclusion. CCI-top1 auditing exposes relative-specificity failures and sparse margin improvement, not broad sign repair. Held-out target response and locality remain separate endpoints; WF superiority is not established.

REFERENCES

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 8748–8763, [Online]. Available: <https://proceedings.mlr.press/v139/radford21a.html>.
- [2] A. Agarwal, S. Karanam, and V. Gandhi, "Concept regions matter: Benchmarking CLIP with a new cluster-importance approach," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2026, pp. 2864–2874, [Online]. Available: <https://arxiv.org/abs/2511.12978>.
- [3] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *The Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000, [Online]. Available: <https://doi.org/10.21314/JOR.2000.038>.
- [4] C. Lin, H. Chen, C. Kim, and S.-I. Lee, "Contrastive corpus attribution for explaining representations," in *International Conference on Learning Representations (ICLR)*, 2023, [Online]. Available: <https://arxiv.org/abs/2210.00107>.
- [5] D. Williamson, Y. Ji, and M. Dwyer, "CASE: Contrastive activation for saliency estimation," *arXiv preprint arXiv:2506.07327*, 2025, [Online]. Available: <https://arxiv.org/abs/2506.07327>.
- [6] Y. Wang, T. G. J. Rudner, and A. G. Wilson, "Visual explanations of image-text representations via multi-modal information bottleneck attribution," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 16 009–16 027, [Online]. Available: <https://doi.org/10.52202/075280-0704>.
- [7] Y. Li, H. Wang, Y. Duan, J. Zhang, and X. Li, "A closer look at the explainability of contrastive language-image pre-training," *Pattern Recognition*, vol. 162, p. 111409, 2025, [Online]. Available: <https://doi.org/10.1016/j.patcog.2025.111409>.
- [8] M. Dreyer, L. Hufe, J. Berend, T. Wiegand, S. Lapuschkin, and W. Samek, "From what to how: Attributing CLIP's latent components reveals unexpected semantic reliance," *arXiv preprint arXiv:2505.20229*, 2025, [Online]. Available: <https://arxiv.org/abs/2505.20229>.
- [9] F. Li, Z. Meng, C. Tan, and Y. Yin, "Class-disentangled attribution with gradient guidance for CLIP," *Journal of King Saud University Computer and Information Sciences*, vol. 38, p. 376, 2026, [Online]. Available: <https://doi.org/10.1007/s44443-026-00779-3>.
- [10] X. Cheng, T. Zhou, S. Yin, H. Zhang, W. Xiong, and W. Ding, "MGA-CLIP: A multigranularity attribution framework for cross-modal explainability in CLIP," *IEEE Transactions on Neural Networks and Learning Systems*, 2026, [Online]. Available: <https://doi.org/10.1109/TNNLS.2026.3696329>.
- [11] R. Visser, I. Roberts, and B. Hammer, "Contrastive concept importance: Explaining pairwise class decisions through automatically extracted concept representations," *arXiv preprint arXiv:2607.27904*, 2026, [Online]. Available: <https://arxiv.org/abs/2607.27904>.
- [12] M. Yuksekgonul, F. Bianchi, P. Kalluri, D. Jurafsky, and J. Zou, "When and why vision-language models behave like bags-of-words, and what to do about it?" in *International Conference on Learning Representations (ICLR)*, 2023, [Online]. Available: <https://openreview.net/forum?id=KRLUvvh8uaX>.
- [13] Z. Ye, F. Jiang, Q. Wang, K. Huang, and J. Huang, "IDEA: Image description enhanced CLIP-adapter for image classification," *Pattern Recognition*, vol. 171, p. 112224, 2026. [Online]. Available: <https://doi.org/10.1016/j.patcog.2025.112224>.
- [14] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollar, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Computer Vision—ECCV 2014*, ser. Lecture Notes in Computer Science, vol. 8693. Springer, 2014, pp. 740–755, [Online]. Available: https://doi.org/10.1007/978-3-319-10602-1_48.
- [15] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL visual object classes (VOC) challenge," *International Journal of Computer Vision*, vol. 88, no. 2, pp. 303–338, 2010, [Online]. Available: <https://doi.org/10.1007/s11263-009-0275-4>.
- [16] T. Gomez, T. Freour, and H. Mouchere, "Metrics for saliency map evaluation of deep learning explanation methods," in *Pattern Recognition and Artificial Intelligence: Third International Conference, ICPRAI 2022*, ser. Lecture Notes in Computer Science. Springer, 2022, pp. 84–95, [Online]. Available: https://doi.org/10.1007/978-3-031-09037-0_8.